Requirements Management

Needs versus wants…

Requirements Process

Wants

vs

Needs

Create

User Stories

Elaborate / Create

New information,

ongoing feedback

Update / Refine

Elicitation

Identify

Stakeholders

User Centered

Design Principles

Prioritise

MoSCoW

Scope

Manage overall

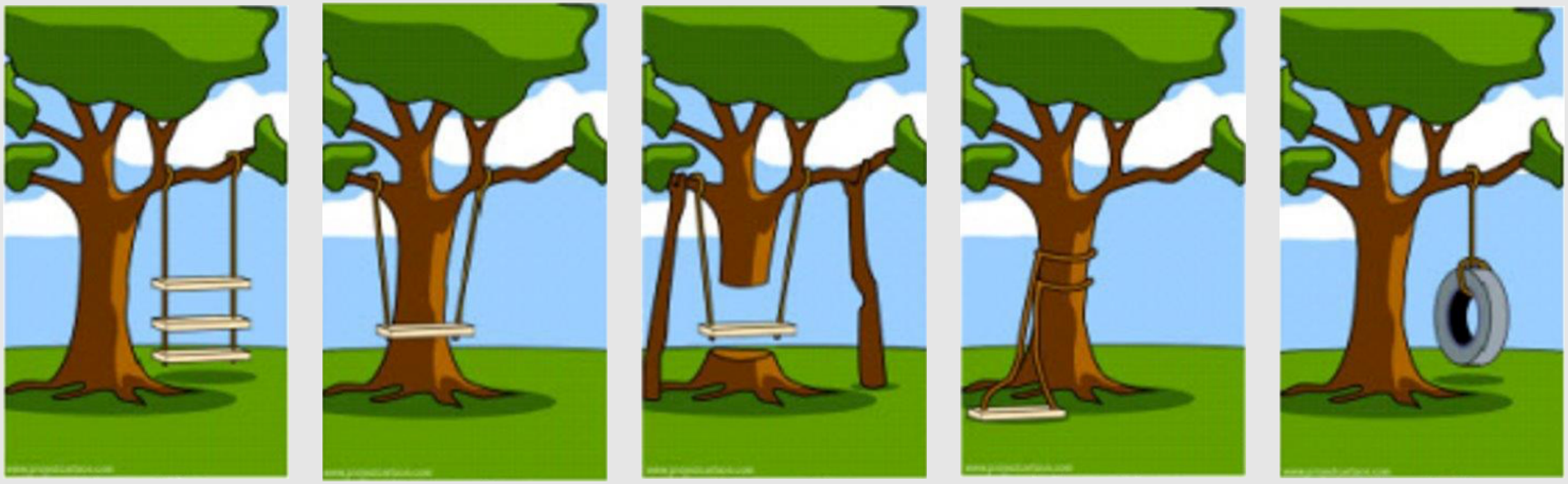
scope and

project direction

Story Mapping

Tanya Matthews @ Version 1

Translating wants to needs



How the customer explained it | How the product owner understood it | How the analyst designed it | How the programmer wrote it | What the customer really needed

User Requirements

Document the process…

Requirements Engineering

- Volere Template is popular
  - Easy to understand by all stakeholders
  - Based on a template
  - Functional Requirements
  - Atomic
  - Measurable
  - Detailed
  - Testable
  - Traceable

Volere Snow Card

Requirement #: Unique ID **Requirement Type:** **Event/BUC/PUC #:** Use Cases ID

Description: The intention of the requirement in one sentence

Rationale: A justification of the requirement i.e why it needs to be part of the product, rather a validation

Originator: Who produced the requirement.

Fit Criterion: A test or measurement of the requirement to verify it meets it's specification

Dependencies: A list of other requirements that have some dependency on this one

Customer Value from two perspectives

Customer Satisfaction: 1-5 **Customer Dissatisfaction:** 1-5

Priority: Some indication of priority **Conflicts:** Other requirements that cannot be implemented if this one is

Supporting Materials: Further documentation on the requirement to support its understanding

History: Creation, changes

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Volere Snow Card

Requirement #: 1 Requirement Type: Funct Event/BUC/PUC #: 1

Description: The product shall allow tutors to send clients automated reminders for payments overdue.

Rationale: Tutors have large numbers of clients and will save time not having to chase up payments.

Originator: Richard Davies

Fit Criterion: When the overdue period has lapsed clients receive a notification about payment.

Customer Satisfaction: 4 Customer Dissatisfaction: 4

Priority: High Conflicts: None

Supporting Materials: None

History: version 1.0

Volere

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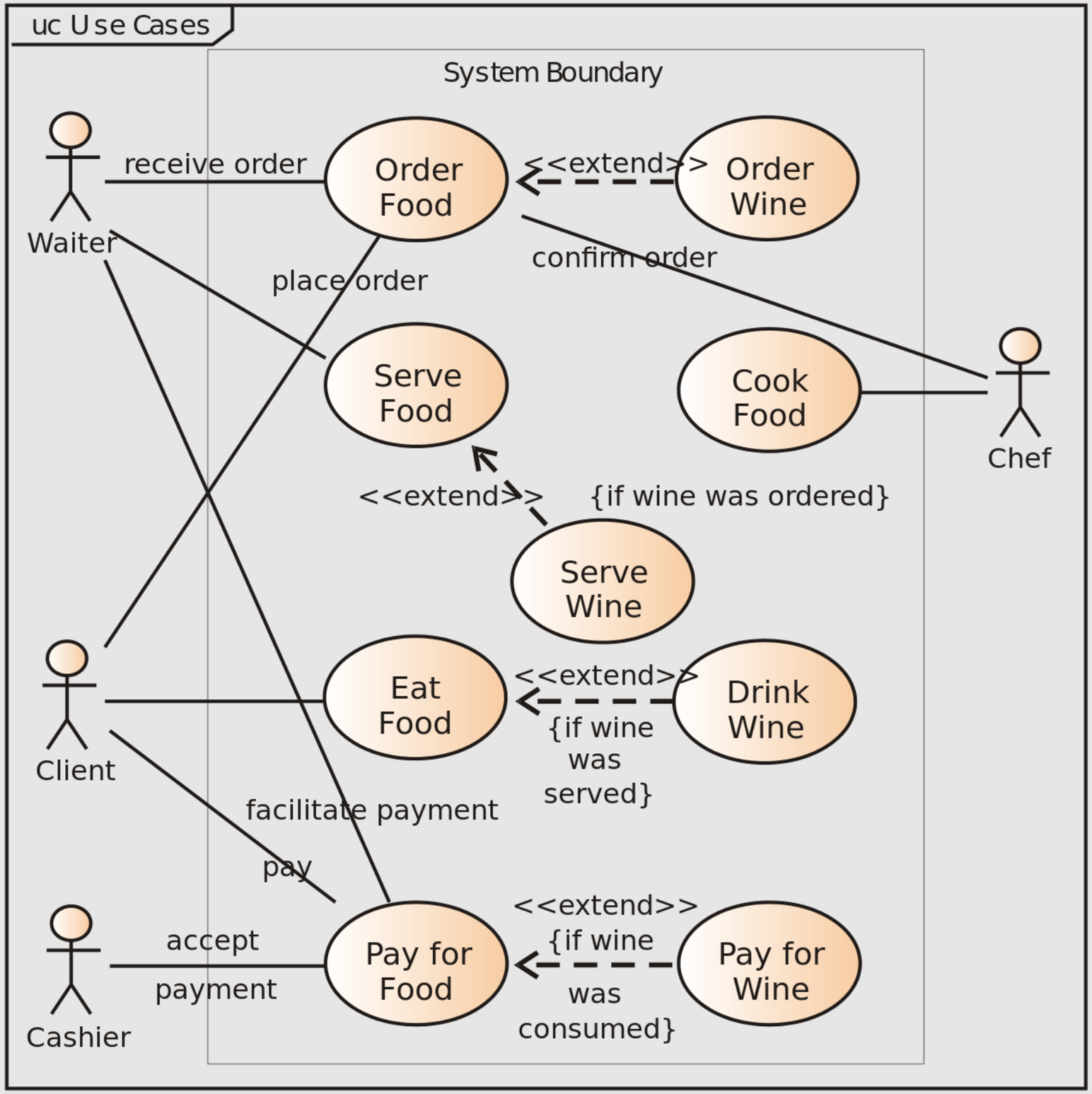
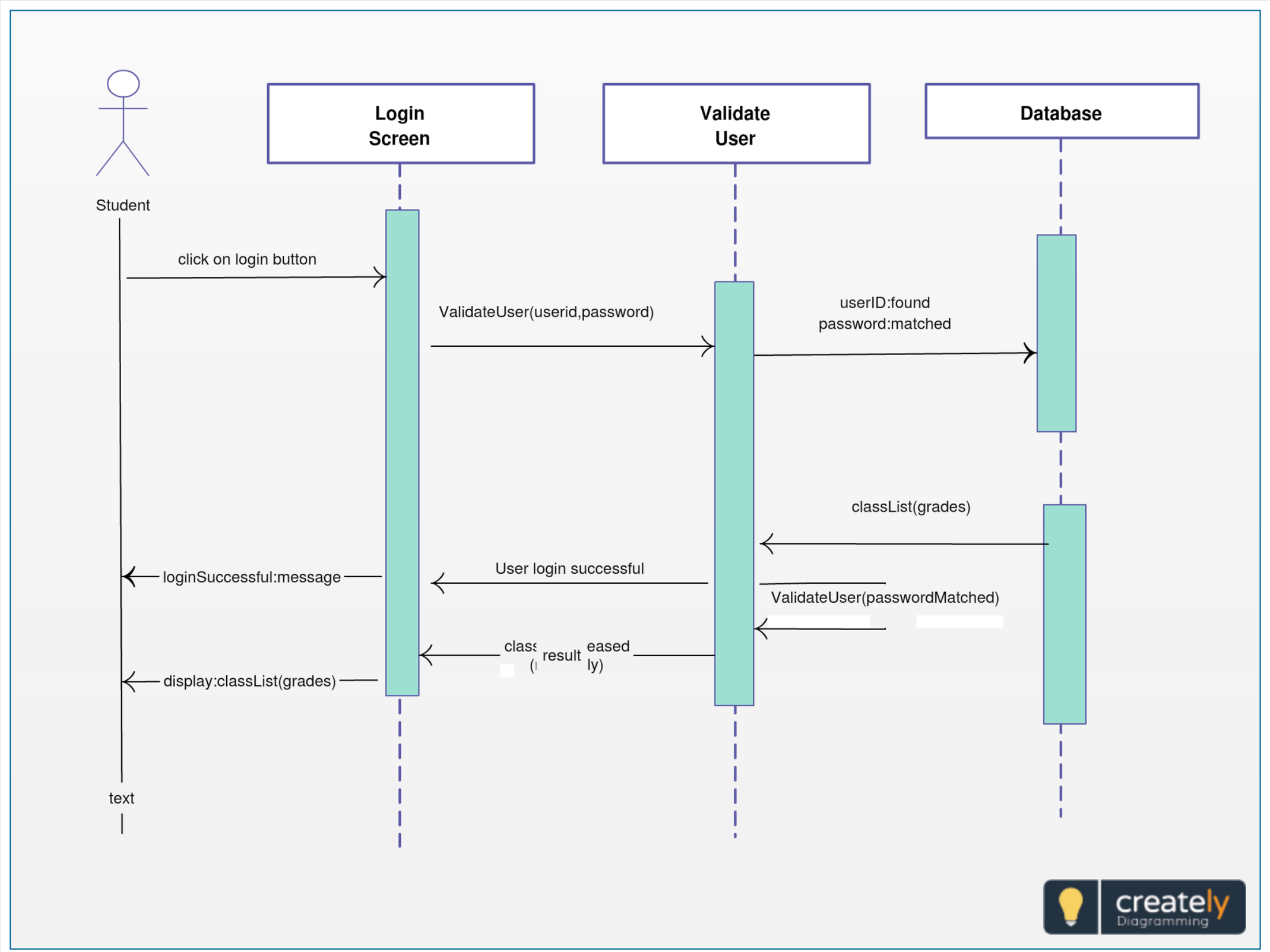
Use Case Diagrams



Sequence Diagrams



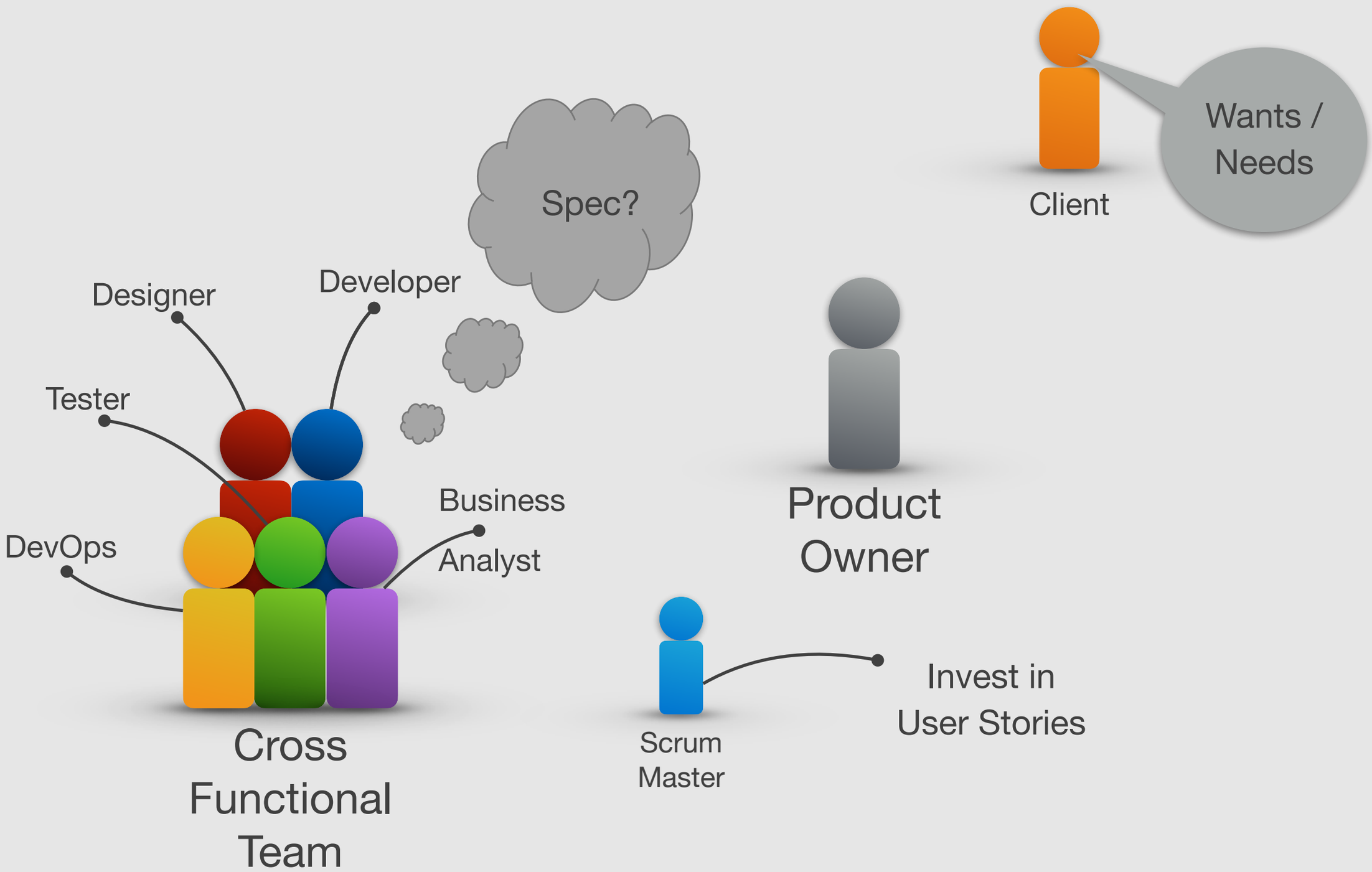
Story Driven Approach

The Agile way…

User Stories

Tell a story from the users perspective…

Who produces User Stories?



INVEST in good user stories

Independent - Self contained to limit dependencies on other user stories

Negotiable - Focus on the what not the how, i.e not driven by the solution

Valuable - Must contain clear user value to be delivered

Estimated - Possible for a team to estimate the stories relative size

Small - Small enough to plan and to complete in 1 or 2 team days (max 1/4 sprint)

Testable - Provide a clear definition of done or acceptance criteria

User Story Format

- Narrative
  - As a **<type of user>**
  - I want to **<do something / have something>**
  - So that **<for some benefit / value is created>**
- Acceptance Criteria (Definition of done)

User Stories

Title: [One line describing an activity]

Narrative: As a role I want to be able to feature so that benefit

Acceptance criteria: [Presented as scenarios]

Scenario structure: Title [that is clearly different from other scenarios]

Given context
And some more context...

When event

Then outcome
And another outcome...

ATM user story - Scenario 1

Title: **Account Holder withdraws cash**

Narrative: As an account holder I want to be able to withdraw cash from an ATM so that I can get money when the bank is closed

Acceptance criteria:

Scenario 1: Account has sufficient funds

Given the account balance is sufficient for the amount requested
And the bank card is valid
And the ATM has sufficient funds

When the account holder requests a cash withdrawal

Then the ATM should dispense the requested amount
And the account balance should be debited by the dispensed amount
And the bank card should be returned

ATM user story - Scenario 2

Title: **Account Holder withdraws cash**

Narrative: As an account holder I want to be able to withdraw cash from an ATM so that I can get money when the bank is closed

Acceptance criteria:

Scenario 2: Account has insufficient funds

Given the account balance is insufficient for the amount requested
And the bank card is valid
And the ATM has sufficient funds

When the account holder requests a cash withdrawal

Then the ATM should inform the customer they have insufficient funds
And ensure that no cash is dispensed
And ensure that the account balance is not adjusted
And the bank card should be returned

ATM user story - Scenario 3….

- ATM has insufficient funds
- Amount exceeds daily limit
- Pin entered incorrectly

ATM user story - In class….

Title: **Account Holder withdraws cash**

Narrative: As an account holder I want to be able to withdraw cash from an ATM so that I can get money when the bank is closed

Acceptance criteria:

Scenario 2: ATM has insufficient funds

Given

And

When

Then

And

Good and bad user stories

As a user, I want to search for tutors ✗

As a tutor, I want to see my daily schedule so I can plan my day and be on time for lessons ✓

As a pupil, I want to see my tutor practicing so that I can better understand how to play ✓

As a parent, I want to monitor lessons, chat with the tutor and pupil via private channels and to record sessions ✗

As a tutor, I want to manage all aspects of payments ✗

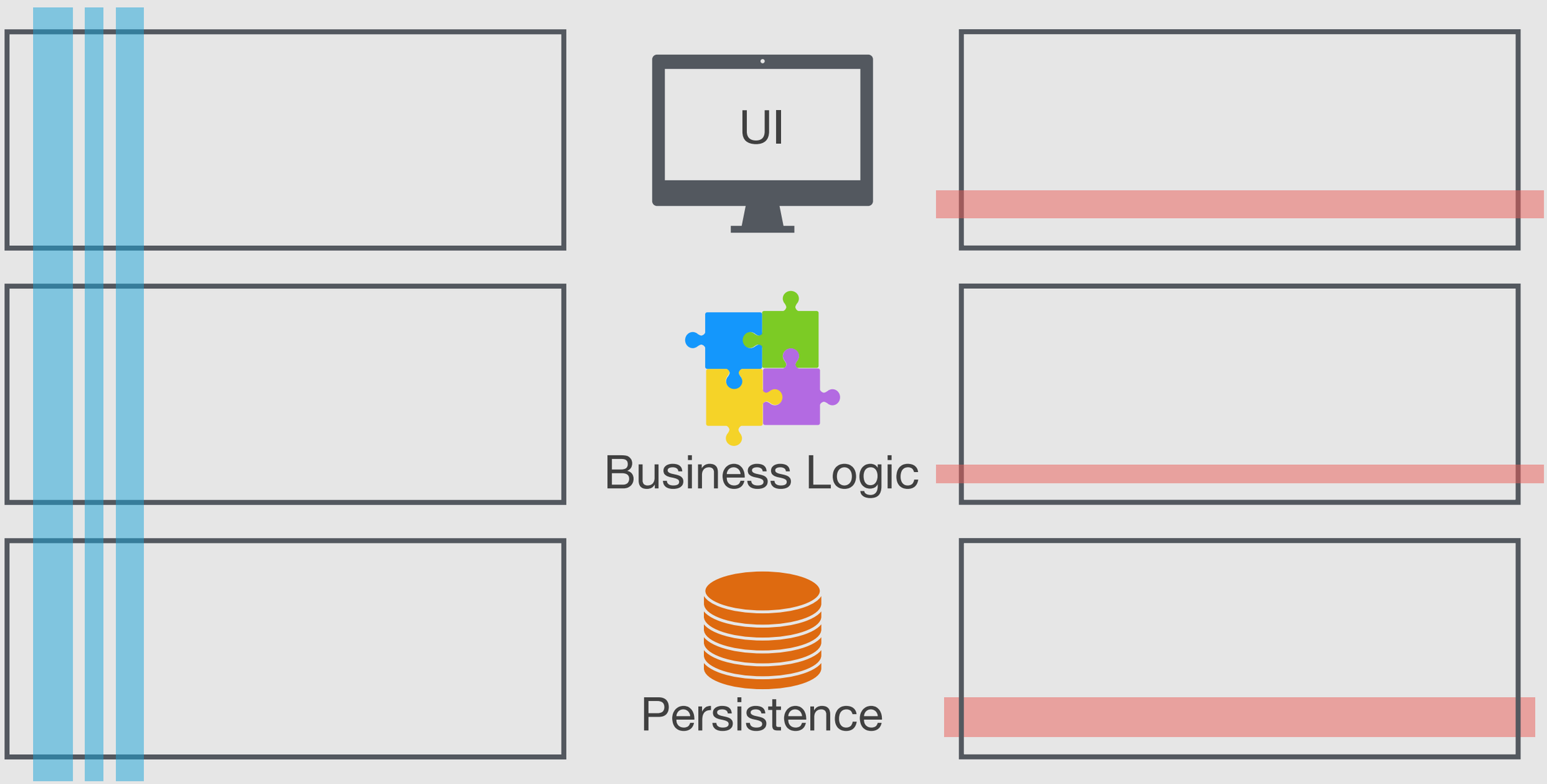
User stories are vertical slices

Vertical Slices

include changes to each architectural layer sufficient to deliver an increment of value

Horizontal Slices

multiple slices must be completed to deliver an increment of value



Acceptance Criteria

- Good AC:
  - Clearly define the boundaries of the story, what's in versus out
  - Describe scenarios that need to be met to satisfy the story as done
  - Use straight forward logical and simple language
  - Scenarios either pass or fail
  - Express as what instead of how, e.g tutor can record video sessions instead of tutors clicks a buttons to start the recording.

Formats of AC

As a user, I want to log-in to my account so that I can view and manage my payment details

Acceptance criteria:

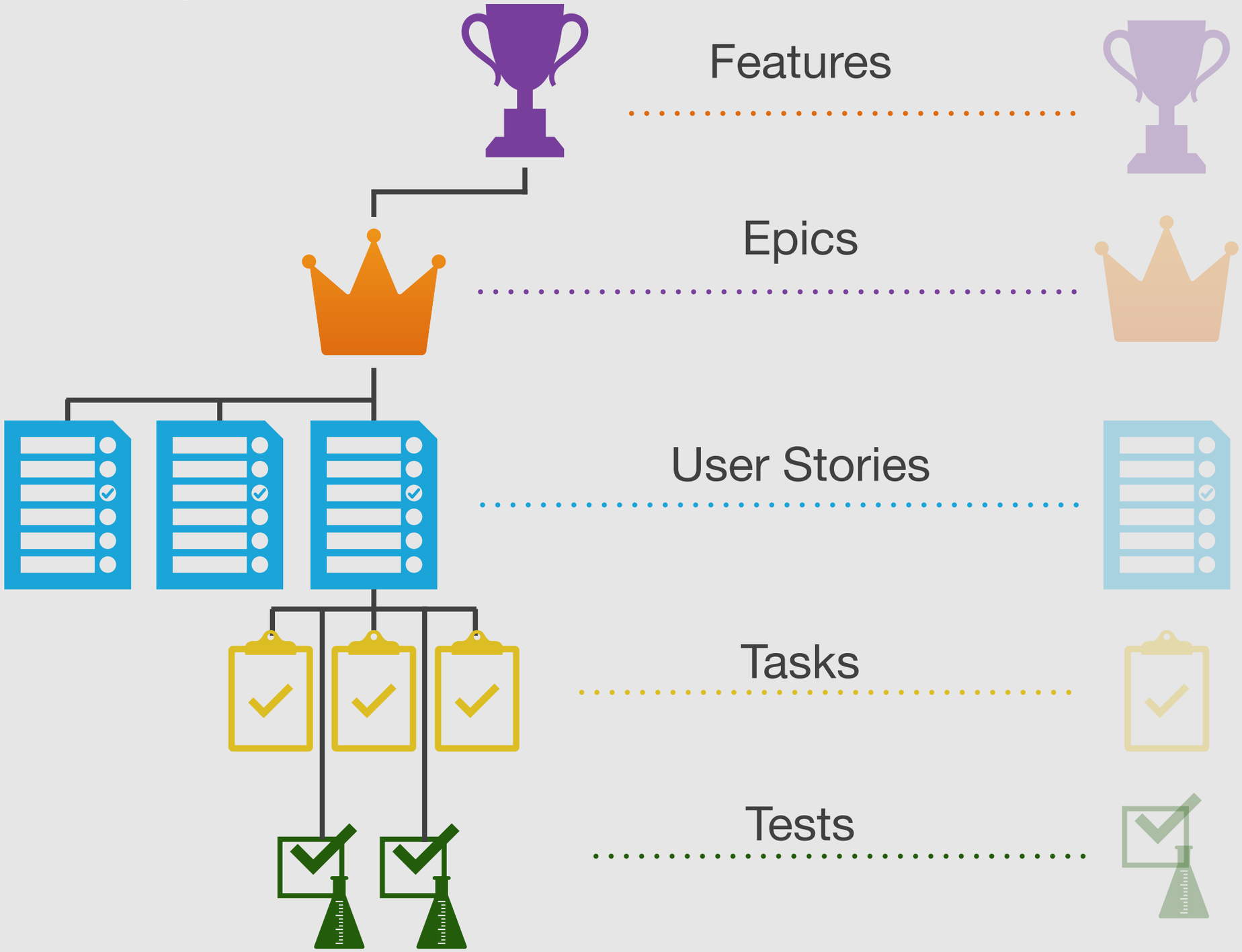
- Given that the user is on the log-in page, when they enter their email address & password, then the system will allow the user to log-in to the system.
- Given that the user has entered their email address and password, when they select to log-in, then the system will show an error message if either the password or email address are invalid.

As a user, I want to log-in to my account so that I can view and manage my payment details

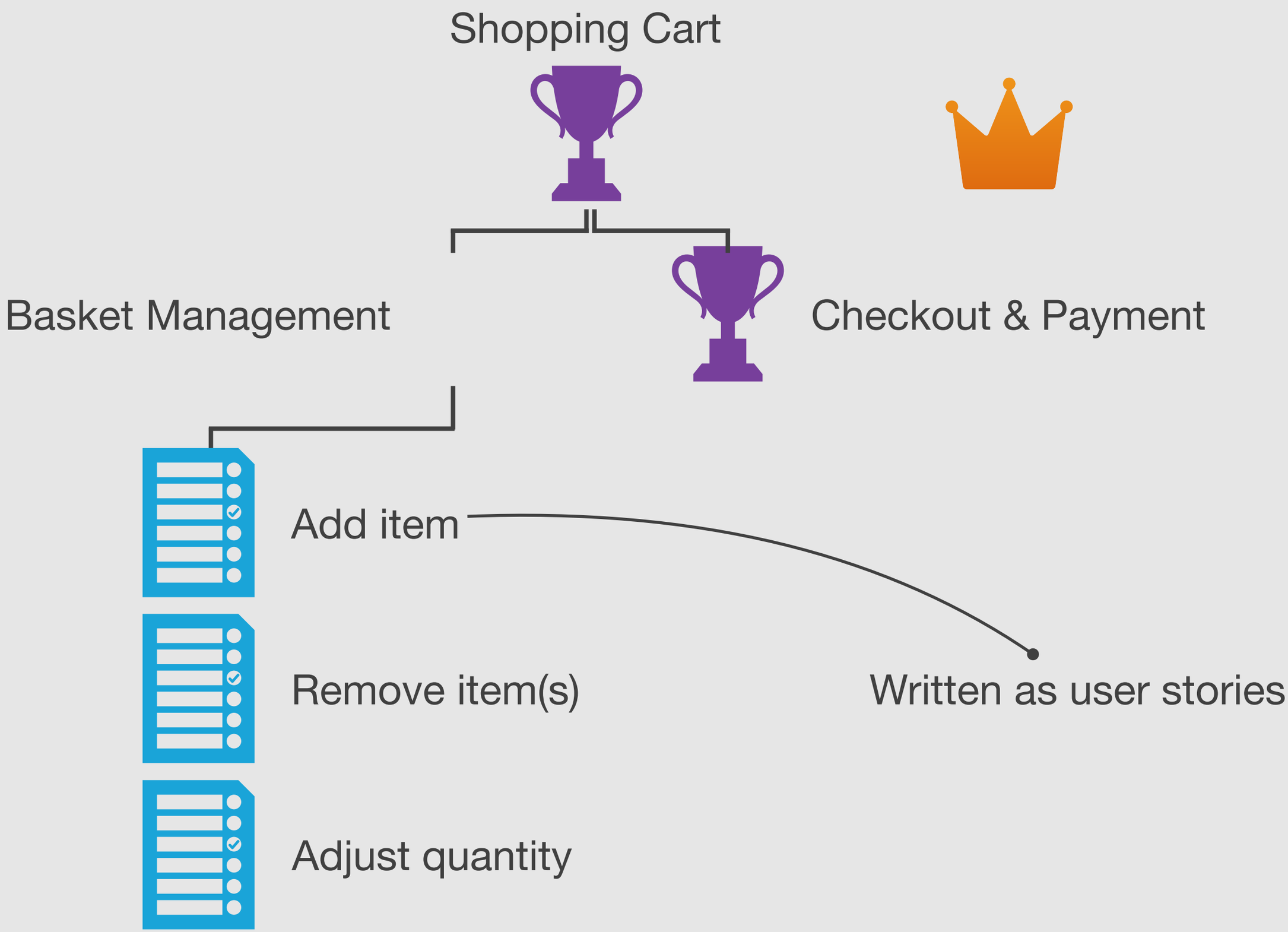
Acceptance criteria:

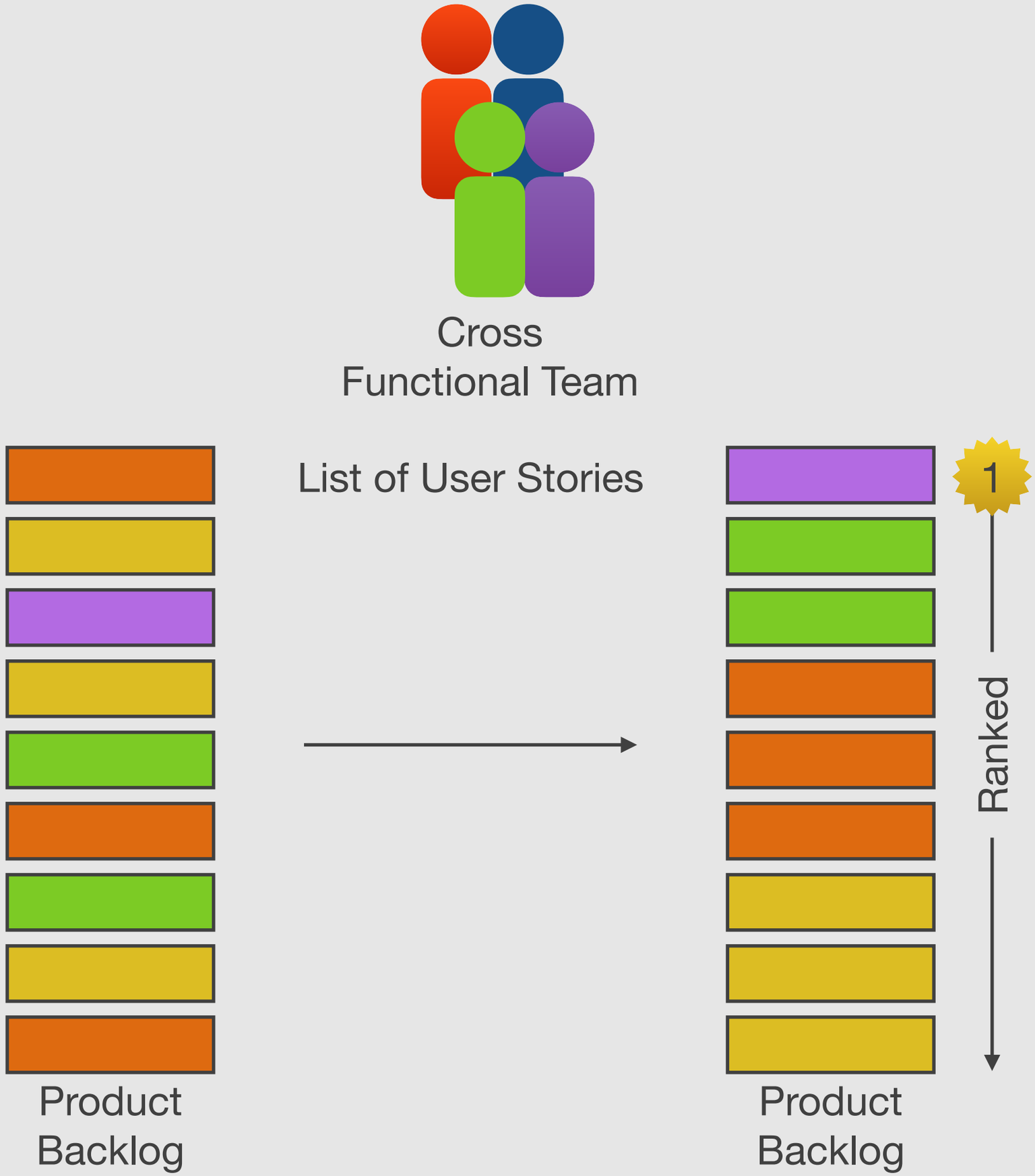
1. Can see the option to log-in
2. Can enter email address & password
3. Can only log-in if both email address & password are valid
4. Can see an error message if either the email address or password are invalid

Hierarchy



Amazon Example




Cross
Functional Team
List of User Stories
1
Ranked
Product
Backlog
Product
Backlog

Influences on priority

Value - Biggest impact on project goals

Not delivering - Biggest negative impact on achieving goals

HiPPO - Highest Paid Persons Opinion

Feedback - Self contained to limit dependencies on other user stories

Difficulty - Level of implementation difficulty

Dependencies - User story A is easier if we do user story B first

Story Mapping

	User Management			Book Room			Update Booking		
	Register	Login	Update Preferences	Search for room	View room info	Book room	View Booking	Change Booking	Cancel Booking
Release 1				By availability	Pictures	Credit card	By booking#		
				By room type	Room details	Email receipt			
Release 2	New User	Sign in		By location				Change date	
								Change room	
Release 3			Update details	3rd party integration		Paypal	By username		Cancel online
			Add payment details						

https://www.productplan.com/glossary/story-mapping/

Story Mapping

- Clearly maps out MVPs
- Identifies gaps / opportunities
- Looks a the big picture
- Helps with prioritisation
- Continuously maps out the product journey